

MC216: Probability and Random Processes – Autumn 2024

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Course Placement

The course is placed in the Third Semester for the B.Tech. MnC program, and is a core course.

Course Format

For the MC216: Probability and Random Processes course, we have (L-T-P-C) = (3-1- 0-4), meaning that it is a **four-credit** course having **three lectures**, and **one tutorial** weekly.

Course Objectives

"This course aims to provide foundational knowledge in probability, random variables, and random processes."

Course Outcomes*

"After completing this course, a student

CO.1 Should be able to understand the basic concepts of probability, random variables, stochastic processes, and a few of their applications in various scientific, technological, and societal applications [PO.1, PO.12, PSO.1].

CO.2 Should have the capability to analyze, comprehend, design & develop engineering processes and systems for a variety of applications and thus demonstrating professional ethics & concern for societal wellbeing [PO.2, PO.3, PO.4, PO.11, PSO.1, PSO.2]."

*For complete program, course, program-specific outcomes, refer to the following link: <https://www.daiict.ac.in/btech-mnc>

Mapping Course Outcomes with Programme Outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	■											■
CO2		■	■	■							■	

Mapping Course Outcomes with Programme Specific Outcomes

	PSO1	PSO2
CO1	■	
CO2	■	■

(Programme Outcomes and Programme Specific Outcomes are given at the end)

Course Overview

In the third semester of the B.Tech. (MnC) programme, students are introduced the fundamentals of probability, random variables, and random processes through this course. These concepts are very essential to the students' professional lives in the next several decades. A familiarity with these ideas is required in every branch of science and engineering, as well as management, economics, business, finance, research, and development, among other fields. For students to gain experience in problem solving, they may be required to write simple codes in a high-level programming language such as MATLAB, Python to solve real-world problems/situations as a part of assignments in the course.

A list of important topics from the course is as follows:

1. Probability: definition and axioms
2. Conditional probability and Bayes' theorem
3. Random variables
4. Discrete random variables, Bernoulli, binomial, and Poisson distributions
5. Continuous random variables, exponential and Gaussian distributions
6. Functions of random variables
7. Mean, variance, moments of random variables
8. Moment generating functions and characteristic functions
9. Joint distributions, covariance and correlation, independent random variables
10. Bi-variate Gaussian distribution
11. Law of large numbers, central limit theorem
12. Random processes and their classification
13. Markov chains, simple random walk
14. Poisson process, Wiener process, white noise, power spectral density
15. Queuing processes

Course Contents*

"Random experiment, sample space, axioms of probability, probability space, conditional probability, independence of events, multiplication rule, total probability rule, Bayes' theorem, random variable, cumulative distribution function, types of random variables, probability mass function, probability density function, distribution of function of random variables, mean and variance, higher order moments and inequalities, generating functions, common discrete distributions, applications of random variables, random vector and joint distribution, joint probability mass function, joint probability density function, independent random variables, functions of several random variables, order statistics, conditional distributions, random sum, moments and co variance, variance- covariance matrix, multivariate normal distribution, probability generating function and moment generating function, correlation coefficient, conditional expectation, modes of convergence, law of large numbers, central limit theorem, stochastic processes, classification of stochastic processes, Bernouli process, Poisson process, simple random walk, time series, strict sense stationary process, wide sense stationary process, discrete time Markov chain, Chapman Kolmogrov equations and N-step transition matrix, classification of states, limiting and stationary distributions, continuous time Markov chain, state transition diagram and Chapman Kolmogrov equation, infinitesimal generator and Kolmogrov differential equations, limiting and stationary distributions, birth-death process, Poisson process, non-homogeneous and compound Poisson process, introduction to queueing models and Kendall notation, M/M/I queueing model, M/M/c queueing model, M/M/I/N model, other Markovian queueing models, transient solution of finite capacity Markovian queues."

*from syllabus for MC216 posted at link:

<https://www.daiict.ac.in/sites/default/files/other-files/Syllabus.pdf>.

Books and Learning Material

- S. Ross, A First Course in Probability, Eighth Edition, Pearson, 2010. [Textbook]
- H. Hsu, Probability, Random Variables and Random Processes, Schaum's Outline, McGraw- Hill Education (India), 2004. [Textbook]
- A. Papoulis, S. Unnikrishna Pillai, Probability, random variables and stochastic processes, New Delhi: Tata McGraw-Hill, 2002 [Reference Book]
- V. Krishnan, Probability and random processes, New Jersey: John Wiley & Sons, 2006 [Reference Book]
- D.C. Montgomery and G. C. Runger, Applied Statistics and Probability for Engineers, 3rd Edition, John Wiley & Sons, 2004
- Papers and documents shared from time to time in Google Classroom/Moodle.

Grading Policy

- Multiple (Surprise) Quizzes – 10%
- Attendance – 10% [(%attendance-80%)/2]
- Multiple (Tutorial) Assignments including group or mini-project assignments – 10%

- Two In-Semester exams – 20% each
- One End-Semester Examination – 30%

Any case of copying/cheating/plagiarism/non-compliance with academic integrity principles/policy abuse, would attract "F" as the final grade irrespective of any other circumstances.

Focus Areas and Details

"Focus on Employability: The students taking the course can be employed in industries focusing on machine learning, data science."

"Focus on Skill Development: The students develop necessary skills to develop foundational knowledge in probability and random processes."

"Focus on Entrepreneurship: –"

Tentative Lecture Schedule/Plan

(Weekly tutorials will follow the course contents being discussed in the class)

Detailed Course Contents		
Topic Name	Content	No. of lectures (typical)
Probability: definition and axioms	<i>Course Introduction, Revisiting various concepts in probability, combinatorial analysis, permutations and combinations, random experiment, sample space, axioms of probability, probability space</i>	2
Conditional probability and Bayes' theorem	<i>conditional probability, independence of events, multiplication rule, total probability rule, Bayes' theorem</i>	2
Random variables	<i>random variable, cumulative distribution function, types of random variables, probability mass function, probability density function, distribution of function of random variables</i>	3
Discrete random variables	Discrete random variables including Bernoulli, binomial, and Poisson and other random variables	2
Functions of Random Variables, Mean and Expected Value	Mean and Expected Value of a Random Variable Mean and Expected Value of a Function of a Random Variable Expected Value of Sums of Random Variable	2
Continuous random variables,	Continuous Random Variables like Uniform, Exponential and Normal/Gaussian distributions Expectation and variance of a continuous random variable Distribution of a function of a continuous random variable.	4
Moment generating functions and	<i>mean and variance, higher order moments and inequalities, generating functions, characteristic functions</i>	3

characteristic functions		
Joint distributions, covariance and correlation, independent random variables	<i>random vector and joint distribution, joint probability mass function, joint probability density function, independent random variables, functions of several random variables, order statistics, conditional distributions, random sum, moments and co-variance, variance-covariance matrix, multivariate normal distribution, probability generating function and moment generating function, correlation coefficient, conditional expectation, bi-variate Gaussian distribution</i>	6
Law of large numbers, central limit theorem	<i>Modes of convergence, Markov, Chebyshev, and other inequalities weak and strong law of large numbers, central limit theorem,</i>	5
Random processes and their classification	<i>stochastic processes, classification of stochastic processes, Bernouli process, Poisson process, Weiner process</i>	3
Discrete-time Markov chains, simple random walk	<i>simple random walk, time series strict sense stationary process, wide sense stationary process, discrete-time Markov chain, Chapman-Kolmogrov equations and N-step transition matrix, classification of states, limiting and stationary distributions,</i>	3
Continuous-time Markov chains	<i>continuous time Markov chain, state transition diagram and Chapman Kolmogrov equation, infinitesimal generator and Kolmogrov differential equations, limiting and stationary distributions, birth-death process, Poisson process, non-homogeneous and compound Poisson process,</i>	3
Queuing processes	<i>Introduction to queueing models and Kendall notation, M/M/I queueing model, M/M/c queueing model, M/M/I/N model, other Markovian queueing models, transient solution of finite capacity Markovian queues.</i>	4

Program Outcomes (POs)

PO No.	Program Outcomes
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the

	engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO No.	Program Specific Outcomes (PSOs)
PSO1	PSO1: To model computational problems by applying mathematical concepts and solving real-world problems using algorithmic techniques.
PSO2	To apply the mathematical and statistical approaches for analyzing, designing and development of computing systems in interdisciplinary applications.